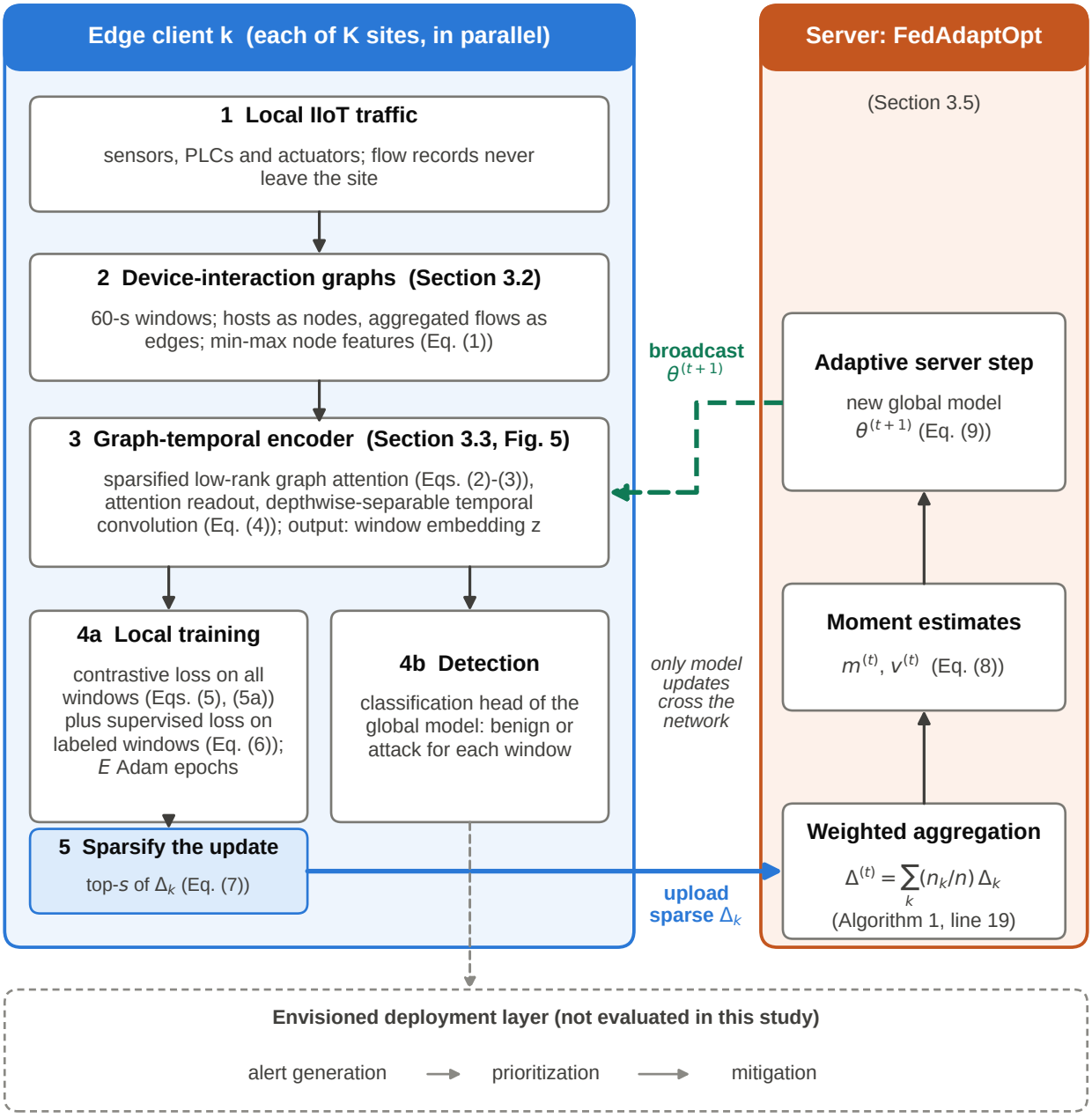
Edge client k (each of K sites, in parallel)
Server: FedAdaptOpt
(Section 3.5)
1 Local IIoT traffic
sensors, PLCs and actuators; flow records never leave the site
2 Device-interaction graphs (Section 3.2)
60-s windows; hosts as nodes, aggregated flows as edges; min-max node features (Eq. (1))
3 Graph-temporal encoder (Section 3.3, Fig. 5)
sparsified low-rank graph attention (Eqs. (2)-(3)), attention readout, depthwise-separable temporal convolution (Eq. (4)); output: window embedding z
4a Local training
contrastive loss on all windows (Eqs. (5), (5a)) plus supervised loss on labeled windows (Eq. (6)); E Adam epochs
4b Detection
classification head of the global model: benign or attack for each window
5 Sparsify the update
top-s of Δk (Eq. (7))
broadcast θ(t+1)
only model updates cross the network
upload sparse Δk
Adaptive server step
new global model θ(t+1) (Eq. (9))
Moment estimates
m(t), v(t) (Eq. (8))
Weighted aggregation
Δ(t) = Σk (nk/n) Δk
(Algorithm 1, line 19)
Envisioned deployment layer (not evaluated in this study)
alert generation
prioritization
mitigation
processing step
client-to-server upload
server-to-client broadcast
envisioned, not evaluated